Recurrence and malignant risk of ameloblastoma: A demographic study of 1626 cases from east China.

PMID: 38061123 | Indexed in PubMed

Ameloblastoma is characterized by aggressive nature, high recurrence rate, occasional malignant transformation, but recurrence and malignant incidence of ameloblastoma are not yet addressed by a large-scale case series study. This study provided a detailed description of the relationship between demographic characteristics and recurrence and malignant cases with different clinical types of ameloblastoma (n = 1626). The overall incidence of recurrence and malignancy was 17.2 % and 3.4 %, respectively. Notably, we observed that there were multiple recurrent episodes (mean time, 24.3-28.7 months) among ameloblastoma patients. Multivariate analysis revealed that age of > 45 years (odds ratios (OR), 2.10; 95 % confidence interval (CI), 1.17-3.76), male (OR, 3.24; 95 %CI, 1.49-6.99), maxilla (OR, 5.58; 95 %CI, 3.11-10.0), and pre-existing recurrence (OR, 3.79; 95 %CI, 2.05-7.01) as independent factors were associated significantly with increased risk of malignancy. Identification of the clinical factors responsible for increased risk of malignancy provides better insight in management planning for ameloblastoma.

Expression of RECK and matrix metalloproteinase-2 in ameloblastoma.

PMID: 19995435 | Indexed in PubMed

Ameloblastoma is a frequent odontogenic benign tumor characterized by local invasiveness, high risk of recurrence and occasional metastasis and malignant transformation. Matrix metalloproteinase-2 (MMP-2) promotes tumor invasion and progression by destroying the extracellular matrix (ECM) and basement membrane. For this proteolytic activity, the endogenous inhibitor is reversion-inducing cysteine rich protein with Kazal motifs (RECK). The aim of this study was to characterize the relationship between RECK and MMP-2 expression and the clinical manifestation of ameloblastoma. Immunohistochemistry and reverse transcription-polymerase chain reaction (RT-PCR) were employed to detect the protein and mRNA expression of RECK and MMP-2 in keratocystic odontogenic tumor (KCOT), ameloblastoma and ameloblastic carcinoma. RECK protein expression was significantly reduced in KCOT (87.5%), ameloblastoma (56.5%) and ameloblastic carcinoma (0%) ($P < 0.01$), and was significantly lower in recurrent ameloblastoma compared with primary ameloblastoma ($P < 0.01$), but did not differ by histological type of ameloblastoma. MMP-2 protein expression was significantly higher in ameloblastoma and ameloblastic carcinoma compared with KCOT ($P < 0.01$). RECK mRNA expression was significantly lower in ameloblastoma than in KCOT ($P < 0.01$), lower in recurrent ameloblastoma than in primary ameloblastoma, and was negative in ameloblastic carcinoma. MMP-2 mRNA expression was significantly higher in ameloblastoma compared with KCOT ($P < 0.01$), but was no different in recurrent ameloblastoma versus primary ameloblastoma. RECK protein expression was negatively associated with MMP-2 protein expression in ameloblastoma ($r = -0.431$, $P < 0.01$). Low or no RECK expression and increased MMP-2 expression may be associated with negative clinical findings in ameloblastoma. RECK may participate in the invasion, recurrence and malignant transformation of ameloblastoma by regulating MMP-2 at the post-transcriptional level.

Enhancing Precision and Esthetics: Endoscopy-Assisted Total Maxillectomy for Locally

Invasive Maxillary Ameloblastoma Using Contralateral Transmaxillary Approach Without Subciliary Incision.

PMID: 39659305 | Indexed in PubMed

This report presents a notable approach to treating a locally invasive maxillary ameloblastoma in a 46-year-old woman using an endoscopy-assisted total maxillectomy via a contralateral transmaxillary approach without a subciliary incision. Ameloblastomas, though benign, require radical surgical management due to their aggressive nature and high recurrence rates, especially in the maxilla. Traditional techniques often involve extensive facial incisions, leading to significant scarring and potential complications. In this case, the notable approach allowed for precise osteotomy of the pterygoid process and accurate delineation of the mucosal margins during ethmoidectomy while preserving facial esthetics. The surgery achieved complete tumor resection with successful reconstruction, avoiding the esthetic drawbacks associated with conventional methods. Postoperative recovery was uneventful, with no residual tumor observed at the six-month follow-up. This technique demonstrates the potential for reducing surgical morbidity and improving cosmetic outcomes in the management of complex maxillary tumors.

Intricacies of plexiform unicystic ameloblastoma: A rare intraluminal journey in the jaw.

PMID: 39547854 | Indexed in PubMed

Ameloblastoma is a true benign odontogenic epithelial tumor, primarily arising in the jaw, and ranks as the second most prevalent odontogenic neoplasm following odontoma. Known for its diverse clinical, radiographic, and histological manifestations, ameloblastoma encompasses a wide spectrum of presentations. Unicystic ameloblastomas (UAs), a less common and generally less aggressive variant, appear as cystic lesions that can mimic ordinary jaw cysts in their clinical and radiologic features. However, histological examination reveals a distinctive odontogenic epithelium lining the cyst cavities, with some cases exhibiting luminal and mural growth. This article presents a unique case involving a 40-year-old female patient who experienced swelling in the right posterior maxilla for four months. Initially presumed to be a routine ameloblastoma, subsequent histopathological analysis identified it as an intraluminal type of UA with rare plexiform changes. It is characterized by a cystic space lined with ameloblast-like cells in a plexiform arrangement, setting it apart from other UA subtypes. Imaging often reveals a unilocular cystic appearance, which may obscure differential diagnosis by closely resembling other odontogenic cysts. The variations within ameloblastoma have always sparked considerable discussion, and we aim to elucidate the reasons behind this specific transformation and its distinctive characteristics.

Ameloblastoma of the Jaw in Three Species of Rodent: a Domestic Brown Rat (Rattus norvegicus), Syrian Hamster (Mesocricetus auratus) and Amargosa Vole (Microtus californicus scirpensis).

PMID: 28942297 | Indexed in PubMed

Ameloblastoma is a locally aggressive tumour derived from the odontogenic epithelium of the developing tooth germ. This uncommon odontogenic tumour is generally considered benign, but rarely, both distant metastasis and cytological atypia occur and this malignant version is referred to as malignant ameloblastic carcinoma. Here we document a spontaneous malignant ameloblastic carcinoma in a rat (Rattus norvegicus) with metastasis to the submandibular lymph node. We also describe ameloblastomas in two other muroid rodents, an Amaragosa vole (Microtus californicus scirpensis) and a Syrian hamster (Mesocricetus auratus). To our knowledge, this is the first report of a malignant ameloblastic carcinoma in any animal and the first report of ameloblastoma in a vole and hamster.

Ameloblastoma of the jaws.

PMID: 10895162 | Indexed in PubMed

Ameloblastoma is a histologically benign tumor derived from odontogenic apparatus. The tumor can infiltrate into surrounding tissues. Although it is benign, it presents symptoms of a malignant tumor, such as infiltration into the lungs, pleura, regional and distant metastases, orbit, base of skull, brain and has resulted in death. It also has a high incidence of recurrences, the existence of regional or distant metastasis, showing a microscopic pattern of ameloblastic carcinoma with cytologic features of an increasing nuclear/cytoplastic ratio, nuclear hyperchromatism, and the presence of mitosis. We report a study of 12 patients of ameloblastoma of the jaws between January 1992 and December 1996 consisting of 8 affected in the mandible and 4 in the maxilla. One patient with a tumor in the maxilla was excluded from this study, due to a different histological and clinical behaviour of the ameloblastoma.

Malignant ameloblastoma: a challenging diagnosis.

PMID: 30775320 | Indexed in PubMed

Ameloblastoma is an uncommon and locally aggressive, benign, odontogenic tumor, with local recurrence when not adequately excised. A rare variant of this neoplasm with the benign features but accompanied with metastases has been described. This rare variant is malignant ameloblastoma and is known to have a poor prognosis. We present the case of a young woman who had recurrent mandibular tumors, which were resected twice and histologically reported as ameloblastoma. Four years later, she presented with pulmonary metastasis and atelectasis. A review of the literature on this very rare neoplasm was also performed.

Malignant ameloblastoma: classification, diagnostic, and therapeutic challenges.

PMID: 11791248 | Indexed in PubMed

Ameloblastoma is a benign odontogenic neoplasm of the mandible and maxilla that rarely exhibits malignant behavior. We report the case of an aggressive malignant ameloblastoma of the mandible that presented with an unusual multiphasic, histologic pattern. Initial fine needle aspiration and radiographic findings showed features consistent with a benign, fibro-osseous lesion. However, aggressive growth and the association of enlarged submandibular lymph nodes suggested a more malignant potential. Treatment consisted of an angle-to-angle composite mandibular resection, right modified neck dissection, left functional supraomohyoid neck dissection, and anterior chin skin resection with iliac crest osteocutaneous free flap reconstruction. Microscopic evaluation showed primarily malignant ameloblastoma without cellular atypia and extensive fields of fibro-osseous tissue with smaller fields of clear cell odontogenic tumor. This multiphasic, histologic arrangement may explain the perplexing preoperative microscopic diagnosis, suggesting a benign fibro-osseous lesion. Of the lymph nodes analyzed, one from the right submandibular triangle exhibited metastatic, benign-appearing ameloblastoma without fibro-osseous or clear cell features. The absence of cellular features of malignancy in the tumor mass and lymph node metastasis suggest that the lesion should be classified as malignant ameloblastoma rather than ameloblastic carcinoma or odontogenic carcinoma. A malignant ameloblastoma with all 3 of the aforementioned microscopic features has not been previously reported. We review the classification of epithelial odontogenic malignancies. Lesions showing multiphasic patterns can create diagnostic dilemmas and may require extensive surgical sampling and/or removal to establish an accurate diagnosis.

Ameloblastic carcinoma of the maxilla.

PMID: 12907214 | Indexed in PubMed

The maxilla is an unusual site for an ameloblastoma, and certainly for an ameloblastic carcinoma. Ameloblastomas are considered as benign, yet locally aggressive neoplasms in the vast majority of cases. However, very rarely, these tumors demonstrate a clinical course of malignancy. Recently, a classification system was published differentiating between malignant odontogenic carcinoma variants. The two such forms include malignant ameloblastoma, and ameloblastic carcinoma. In the differential diagnosis, the designation of malignant ameloblastoma is reserved for lesions that, despite their benign histology, metastasize as well-differentiated cells. The diagnosis of ameloblastic carcinoma is reserved for tumors that demonstrate a malignant morphologic appearance, regardless of whether metastasis is a proven fact at the time of discovery and treatment. We discuss the presentation, pathology, and treatment of the 18th case of a maxillary ameloblastic carcinoma in the literature.

The variability and complexity of ameloblastoma: carcinoma ex ameloblastoma or primary ameloblastic carcinoma.

PMID: 22920533 | Indexed in PubMed

Ameloblastoma is characterized by slow-growing, local invasiveness and high incidence of local recurrence. It usually presents with a benign histological appearance. However, ameloblastoma occasionally demonstrates a clinical course that is characteristic of malignant transformation. Here, we present a case of ameloblastoma with an aggressive clinical course, including multiple recurrences, a short disease-free interval, pulmonary metastasis and extensive skull-base infiltration. With a careful re-evaluation of the histology and cytology of the specimens of primary and recurrent ameloblastoma in 2006 and 2007, malignant transformation was observed and carcinoma ex ameloblastoma was ultimately diagnosed.